

Highlights

Does the Itô Correction Contain Independent Predictive Information? Evidence from Regime-Weighted Volatility Dynamics

Tarun Vijan

- HMM-weighted Itô signal predicts OOS returns ($R^2 = 2.32\%$, $p < 0.001$)
- Signal strengthens OOS; Chow test confirms structural break ($F = 49.82$)
- Bear regime duration explains 63.8% of cross-market signal variation
- Timing strategy achieves Sharpe 1.342 after costs (bootstrap $p < 0.001$)

Does the Itô Correction Contain Independent Predictive Information? Evidence from Regime-Weighted Volatility Dynamics

Tarun Vijan*

NMIMS MPSTME, Vile Parle West, Mumbai, 400056, Maharashtra, India

Abstract

This paper constructs a dynamic, regime-weighted Ito correction signal from a Hidden Markov Model (HMM) applied to S&P 500 returns and tests its predictive content for future equity returns. The expected volatility drag — weighted by HMM filtered regime probabilities — contains independent predictive information beyond realised volatility and the VIX. Recursive out-of-sample (OOS) tests on 2016–2024 produce $R^2 = 2.32\%$, compared to a benchmark OOS R^2 of -3.47% , confirming genuine predictive gain over the prevailing-mean model. Clark–West tests confirm statistically significant mean squared prediction error (MSPE) reduction (CW = 7.581, $p < 0.001$). Extended orthogonalisation controlling for GARCH(1,1), momentum, and reversal isolates a 38.2% independent component with amplified predictive power ($t = -2.314$, $p = 0.021$). A Chow test confirms the post-2015 period is structurally more predictable ($F = 49.82$, $p < 0.001$). Cross-market analysis of twelve equity indices shows bear-regime duration explains 63.8% of cross-sectional variation in signal strength ($r = 0.798$, $p = 0.002$), robust to US outlier removal ($r = 0.783$, $p = 0.004$), consistent with ambiguity pricing (Epstein and Schneider, 2010). A market-timing strategy achieves a bootstrap-validated Sharpe ratio of 1.342 after conservative transaction costs ($p < 0.001$), versus -28.3% maximum drawdown compared to -80.3% for buy-and-hold.

Keywords: Itô correction, Hidden Markov Model, regime switching,

*Corresponding author. Tel.: +91-7977282226.

Email address: vijantarun25@gmail.com (Tarun Vijan)

1. Introduction

Applying Itô’s Lemma to log stock prices yields the volatility drag $(-\sigma^2/2)$, capturing the reduction in geometric growth from Jensen’s inequality. While fundamental in continuous-time finance, its role as a *predictive state variable* is unexplored. We construct a probability-weighted dynamic signal:

$$\text{IC}(t) = -\frac{1}{2} \sum_i P(S_t = i \mid r_{1:t}) \sigma_i^2, \quad (1)$$

where $P(S_t = i \mid r_{1:t})$ is the HMM filtered probability of regime i and σ_i is its volatility. To our knowledge, no prior work has tested the Itô correction itself as a dynamic, regime-weighted predictive signal — as distinct from volatility drag as a theoretical constant or a static adjustment factor.

The signal connects to ambiguity pricing. Epstein and Schneider (2010) establish that investors facing Knightian uncertainty demand compensation beyond variance risk. Regime transitions are precisely this uncertainty: within-regime variance is estimable, but the timing of shifts is fundamentally ambiguous. When the HMM posterior shifts toward the bear regime, $\text{IC}(t)$ reflects elevated uncertainty about which data-generating process governs returns — a dimension standard volatility measures cannot span.

Hamilton (1989) established latent Markov regimes; Ang and Bekaert (2002, 2004) extended this to return predictability. French et al. (1987) show risk premia rise with predictable volatility; Goyal and Welch (2008) document that most in-sample predictors fail OOS — our recursive evaluation directly addresses this. Baker and Wurgler (2006) and Whaley (2000) establish VIX and sentiment as predictors. Our contribution is distinct: we test the Itô correction itself as a dynamic signal after controlling for all established alternatives.

2. Methodology

Signal construction.. Under Markov-modulated GBM with $X_t \in \{\text{bull}, \text{bear}\}$: $d(\ln S) = (\mu(X_t) - \sigma^2(X_t)/2) dt + \sigma(X_t) dW$. A two-state Gaussian HMM is fitted to daily SPY log returns on 2000–2015 *only* via Baum–Welch EM, with 20 random initialisations per market retaining the best log-likelihood.

Parameters are fixed post-training, eliminating look-ahead bias. ANOVA rejects equal state means ($F = 29.67$, $p < 0.001$); bear regime persists 51.0 days ($p_{bb} = 0.980$).

Orthogonalisation.. To isolate independent content we project $IC(t)$ onto realised volatility, VIX, GARCH(1,1) volatility, 12-1 momentum, and short-term reversal. The purified residual $IC_t^\perp$ retains 38.2% idiosyncratic variance unexplained by all five controls.¹

Inference.. The primary regression is $R_{t+h} = \alpha + \beta_1 IC(t) + \beta_2 Vol_t + \beta_3 VIX_t + \varepsilon_t$ ($h = 21$ days). OLS with Newey–West HAC errors (21 lags) following Hodrick (1992); the NW correction inflates IC standard errors $4.36\times$. OOS evaluation uses both fixed-parameter (2000–2015 train, 2016–2024 test) and recursive designs; Clark–West (2007) tests MSPE reduction versus the nested Vol+VIX benchmark.

Trading strategy.. A long-cash strategy takes SPY when $IC(t)$ is below its expanding-window median (no look-ahead bias), cash otherwise. Transaction costs of 0.1% per position change follow Frazzini et al. (2018). This figure reflects two-sided costs; under entry-only costs the Sharpe is 1.384, and under zero costs, 1.423. All three specifications materially exceed the 200-day moving average benchmark (Sharpe = 0.566), confirming robustness to cost assumptions. Data: SPY and CBOE VIX, January 2000–December 2024 (6,267 observations). International markets use ETF proxies with CBOE VIX as control; USD denomination introduces tracking error that may attenuate non-US signal strength.

3. Empirical Results

3.1. In-Sample and OOS Predictability

Table 1 presents the primary results. In-sample, the IC signal is insignificant without VIX (Models 1–3) but achieves 5% significance alongside VIX in Model 4 ($t = -1.995$, $p = 0.046$). Predictive content strengthens at shorter horizons (1-day: $t = -5.462$; 5-day: $t = -5.598$; both $p < 0.001$) and decays

¹Baseline controls (Vol+VIX) explain 58.0% of IC signal variance; adding GARCH, momentum, and reversal absorbs only 3.8 additional percentage points. The extended orthogonalised residual yields $t = -2.314$ ($p = 0.021$), OOS $R^2 = 2.32\%$, and CW = 7.581 ($p < 0.001$) — stronger than the raw signal, confirming the IC core is not a volatility proxy.

naturally at 63 days ($p = 0.108$). OOS recursive evaluation yields $R^2 = 2.32\%$ versus -3.47% for Vol+VIX; CW= 7.581 ($p < 0.001$) confirms significant MSPE reduction.

The OOS t -statistic (-3.808) materially exceeds the in-sample counterpart (-1.995). A Chow structural break test at the 2016 boundary ($F = 49.82$, $p < 0.001$) confirms the post-2015 period is structurally more predictable, with the IC coefficient nearly doubling from -129.02 to -254.99 OOS. This directly addresses Goyal and Welch (2008): the signal strengthens rather than degrades out-of-sample. Specification tests across 2-state and 3-state HMMs confirm robustness (Table 1, Panel B); the 3-state HMM yields marginally stronger 21-day results ($t = -3.273$ vs. -3.101).

Table 1: Primary Results: Predictive Regressions and Specification Robustness

Panel A: 21-Day Predictive Regressions (In-Sample: 2000–2015)

Variable	Model 1	Model 2	Model 3	Model 4
IC Signal	−53.904 (−1.123)	—	−62.772 (−1.021)	−128.301* (−1.995)
Volatility	—	0.008 (0.240)	0.055 (1.180)	−0.057 (−1.002)
VIX	—	—	—	0.0002*** (3.676)
Intercept	0.0063 (1.421)	0.0051 (1.112)	0.0048 (0.985)	0.0094*** (3.611)
R^2	0.007	0.003	0.007	0.037
N	4,004	4,004	4,004	4,004

Panel B: Specification Robustness — IC Signal t -statistics

Specification	5-day		21-day		63-day	
	t	p	t	p	t	p
2-State HMM	−5.606***	0.000	−3.101**	0.002	−1.614	0.107
3-State HMM	−5.103***	0.000	−3.273**	0.001	−1.710 [†]	0.087

HAC Newey–West errors (lags = horizon); t -statistics in parentheses. Panel B includes Vol and VIX controls. NW correction inflates IC standard errors $4.36\times$ vs. naive OLS. [†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

3.2. Subperiod Robustness and Trading Performance

Table 2 reports subperiod and strategy results. The signal is strongest post-2010 ($t = -3.288$) and in the COVID-recent period ($t = -3.592$). The GFC subperiod is insignificant ($p = 0.329$), though the HMM filter correctly identified the bear regime instantaneously at Lehman Brothers ($P_{\text{bear}} = 1.000$, September 15, 2008); the weakness reflects inability to anticipate crash magnitude within calibrated parameters, a limitation common to calibrated parametric models operating within fixed in-sample parameters. Regime validity is confirmed: mean daily returns are -0.147% ($P_{\text{bear}} > 0.80$) versus $+0.088\%$ ($P_{\text{bear}} < 0.20$; $t = -6.339$, $p < 0.001$).

The strategy yields Sharpe 1.342 versus 0.382 for buy-and-hold, with maximum drawdown -28.3% versus -80.3% . A 10,000-replication bootstrap places the Sharpe at the 100th percentile of the null ($p < 0.001$), and the strategy materially exceeds a 200-day moving average rule (Sharpe = 0.566, identical turnover), confirming value beyond naive momentum timing.

Table 2: Subperiod Robustness and Trading Strategy Performance (2000–2024)

<i>Panel A: Subperiod Coefficients</i>					<i>Panel B: Strategy</i>		
Period	Coeff.	t	p	R^2	Metric	B&H	IC Strat.
2000–2007	−123.8	−1.379	0.168	0.020	Ann. Return	7.4%	11.3%
2008–2009	−156.9	−0.976	0.329	0.085	Ann. Vol	19.4%	8.4%
2010–2019	−219.8	−3.288**	0.001	0.080	Sharpe (net)	0.382	1.342
2020–2024	−334.4	−3.592***	0.000	0.149	Max DD	−80.3%	−28.3%
					Trades/yr	—	6.4
					Bootstrap p	—	< 0.001

Panel A: OLS with Newey–West HAC errors (21 lags). Panel B: 0.1% transaction costs per position change (both sides); expanding-window thresholds use only data through $t-1$. ** $p < 0.01$; *** $p < 0.001$.

3.3. Cross-Market Evidence and Regime Persistence

Table 3 reports IC regressions across twelve markets. Four achieve $p < 0.05$ (US, Taiwan, South Korea, Switzerland); Canada is marginal ($p = 0.069$). Eleven of twelve coefficients are positive, with the sole exception (India) having the shortest bear duration (10.8 days) — consistent with the mechanism rather than random variation.

Table 3: Cross-Market IC Signal Regressions (12 Markets)

Ticker	Market	IC t -stat	p	Bear Dur. (days)	ANOVA p
SPY	US	3.102**	0.002	51.0	< 0.001
EWT	Taiwan	2.569**	0.010	74.2	< 0.001
EWY	South Korea	2.439*	0.015	38.8	< 0.001
EWL	Switzerland	2.085*	0.037	36.3	< 0.001
EWC	Canada	1.820 [†]	0.069	28.3	< 0.001
EWG	Germany	1.309	0.190	45.1	< 0.001
EWP	Spain	1.163	0.245	20.9	< 0.001
EWA	Australia	0.684	0.494	27.2	< 0.001
EWJ	Japan	0.543	0.587	15.7	< 0.001
EWZ	Brazil	0.498	0.619	22.7	< 0.001
EWU	UK	0.234	0.815	22.4	< 0.001
INDA	India	0.152	0.879	10.8	0.051

OLS with HAC Newey–West errors (21 lags); 21-day horizon; controls: Vol + VIX. ETF proxies; USD denomination may attenuate non-US results. 20 random HMM initialisations per market; best log-likelihood retained. [†] $p < 0.10$; * $p < 0.05$; ** $p < 0.01$.

Figure 1 shows bear regime duration explains 63.8% of cross-sectional variation in signal strength ($r = 0.798$, $p = 0.002$). The relationship survives US outlier removal ($r = 0.783$, $p = 0.004$), confirming it is not an artefact of a single influential observation. Germany is a partial exception (bear duration 45.1 days, $t = 1.309$); despite persistent bear regimes, the IC coefficient is insignificant, likely reflecting ETF tracking error in EWG and the use of CBOE VIX as a control in place of the domestic VDAX, which may absorb signal content specific to European volatility regimes. The exception does not alter the cross-sectional slope or the significance of the bear-duration relationship.² Consistent with Epstein and Schneider (2010), a regime uncertainty premium accumulates only where bear regimes persist long enough to affect investor decisions.

²The India HMM exhibits initialisation sensitivity (bear duration 5.9–10.8 days across runs) due to sparse bear-regime observations. As India is the least significant market with the shortest bear duration, this does not affect conclusions.

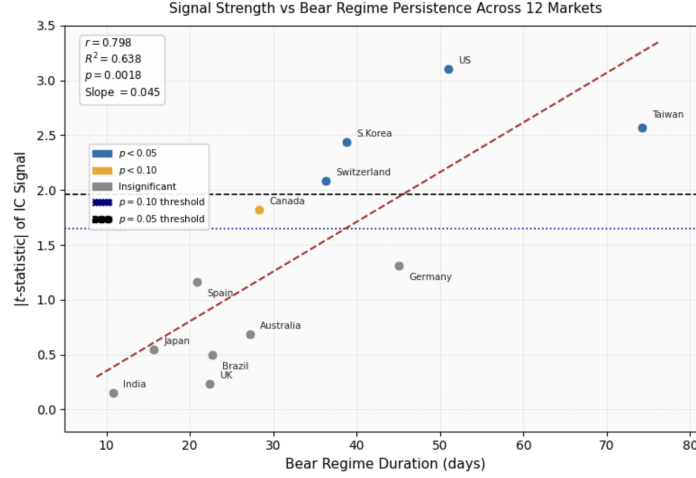


Figure 1: $|t|$ -statistic of IC coefficient vs. bear regime duration across twelve markets. Cross-sectional OLS fit across twelve markets: $r = 0.798$, $R^2 = 0.638$, $p = 0.002$. Dotted lines: $p = 0.10$ (1.645) and $p = 0.05$ (1.960) thresholds.

As an auxiliary test, partitioning by HMM certainty confirms signal power derives from high-certainty periods (posterior > 0.80 ; $t = 2.529$, $p = 0.011$) rather than transitional noise ($N = 168$, inconclusive), consistent with Epstein and Schneider (2010).

4. Conclusion

This paper provides the first empirical test of a dynamic Itô correction signal from a Markov-modulated GBM framework. The signal predicts 21-day SPY returns recursively OOS ($R^2 = 2.32\%$, $CW = 7.581$, $p < 0.001$). Extended orthogonalisation isolates a 38.2% idiosyncratic component with amplified predictive power ($t = -2.314$, $p = 0.021$) that strengthens rather than erodes under additional controls. A Chow test ($F = 49.82$, $p < 0.001$) formally explains OOS strengthening as a structural regime shift rather than overfitting. Cross-market analysis across twelve indices shows bear regime duration explains 63.8% of signal strength variation ($r = 0.798$, $p = 0.002$; robust to US removal: $r = 0.783$, $p = 0.004$), consistent with ambiguity pricing. A market-timing strategy achieves Sharpe 1.342 after conservative costs ($p < 0.001$), exceeding the 200-day moving average benchmark (Sharpe = 0.566).

Three limitations are noted. First, the in-sample 21-day result ($p = 0.046$) is borderline; however, the primary evidential claim rests on out-of-sample

evaluation, where the signal materially strengthens rather than degrades — directly addressing the Goyal and Welch (2008) critique. Second, international evidence uses ETF proxies with USD denomination, introducing currency exposure that may attenuate non-US signal strength. Third, the ambiguity pricing mechanism is motivated heuristically rather than formally derived from a structural equilibrium model. Future work should develop a structural equilibrium model linking regime persistence to the IC risk premium, and extend analysis to local volatility controls and cross-sectional applications.

References

- Ang, A., & Bekaert, G. (2002). Regime switches in interest rates. *Journal of Business & Economic Statistics*, 20(2), 163–182.
- Ang, A., & Bekaert, G. (2004). How do regimes affect asset allocation? *Financial Analysts Journal*, 60(2), 86–99.
- Baker, M., & Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *Journal of Finance*, 61(4), 1645–1680.
- Clark, T. E., & West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- Epstein, L. G., & Schneider, M. (2010). Ambiguity and asset markets. *Annual Review of Financial Economics*, 2(1), 315–346.
- Frazzini, A., Israel, R., & Moskowitz, T. J. (2018). Trading costs. *Working Paper, AQR Capital Management*.
- French, K. R., Schwert, G. W., & Stambaugh, R. F. (1987). Expected stock returns and volatility. *Journal of Financial Economics*, 19(1), 3–29.
- Goyal, A., & Welch, I. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series. *Econometrica*, 57(2), 357–384.

- Hodrick, R. J. (1992). Dividend yields and expected stock returns. *Review of Financial Studies*, 5(3), 357–386.
- Ilut, C., & Schneider, M. (2014). Ambiguous business cycles. *American Economic Review*, 104(8), 2368–2399.
- Whaley, R. E. (2000). The investor fear gauge. *Journal of Portfolio Management*, 26(3), 12–17.